

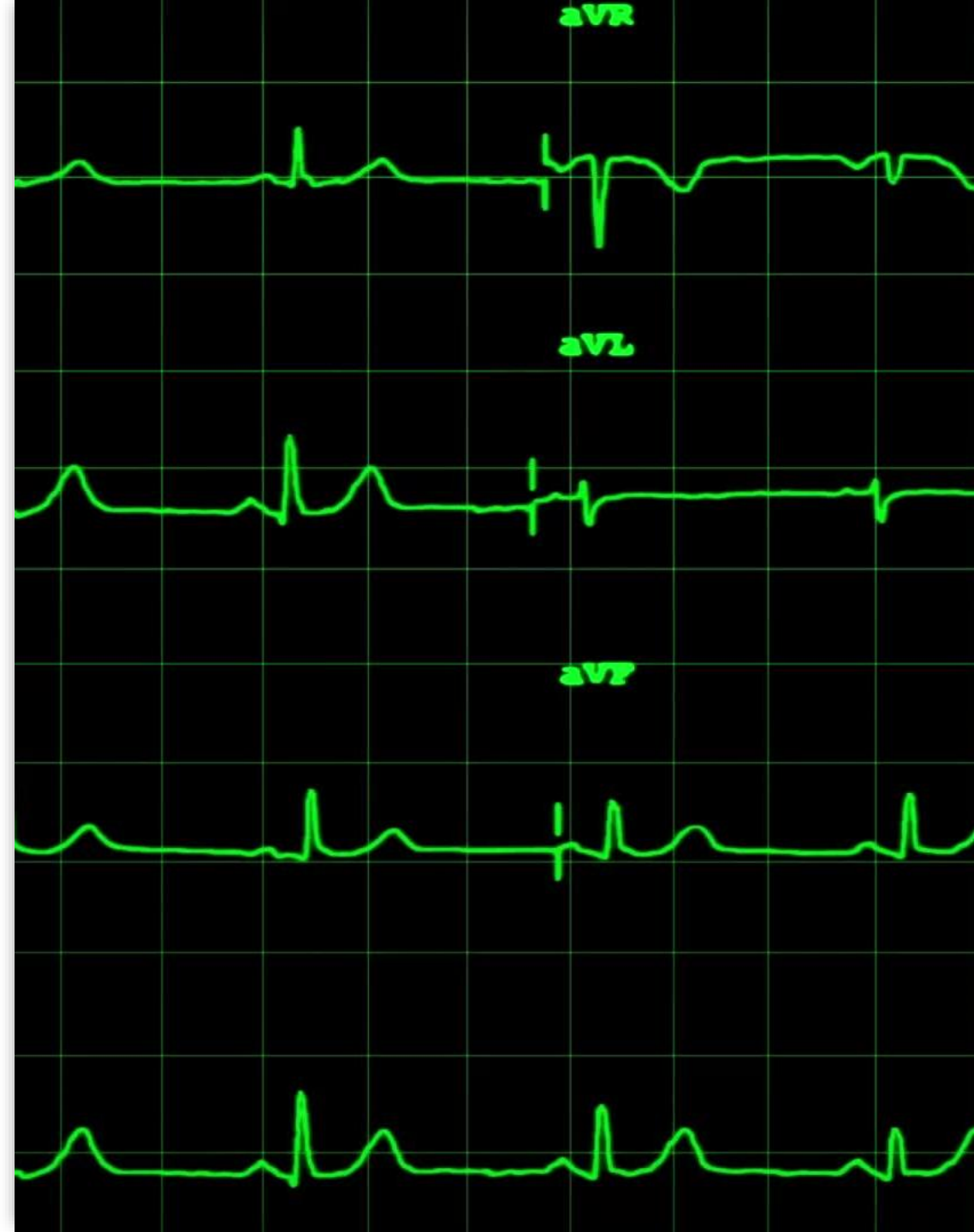


Terminology

Examples of Diseases: Their Signs
and Symptoms

Learning Outcomes of the Lecture

- By the end of this lecture, students should be able to:
 1. **Define** the terms *disease*, *sign*, and *symptom*.
 2. **Differentiate** between a *sign* and a *symptom* using clear clinical examples.
 3. **List** common signs and symptoms of selected common diseases (e.g. pneumonia, myocardial infarction, diabetes, anemia, appendicitis, stroke, heart failure, cirrhosis, UTI).
 4. **Relate** groups of signs and symptoms to the most likely affected body system.
 5. **Use correct medical terminology** to describe signs and symptoms when presenting a simple clinical case.



Definition:

Disease:

A disturbance of normal structure or function of the body that produces specific signs and/or symptoms.

Symptom:

A *subjective* complaint described by the patient (cannot be directly measured).

- Example: headache, nausea, “I feel tired”, pain.

Sign:

An *objective* finding observed by the doctor or measured by tests.

- Example: fever (38.5°C), rash, high blood pressure, jaundice.

Difference Between Sign and Symptom

Aspect	Symptom (Subjective)	Sign (Objective)
Who notices?	Patient	Doctor / nurse / observer
Can we measure?	Usually no	Usually yes (exam or investigation)
Example 1	Chest pain	ST elevation on ECG, low blood pressure
Example 2	Shortness of breath	Respiratory rate 30/min, low oxygen saturation
Example 3	Feeling hot	Temperature 39°C

Why Signs and Symptoms Are Important

Help to:

1. Detect disease early
2. Localize the affected organ or system
3. Decide on investigations and treatment
4. Communicate clearly between healthcare professionals



Respiratory

Example:

Pneumonia

Symptoms (patient complains of):

- 1) Cough (may be productive with sputum)
- 2) Chest pain, especially on breathing (pleuritic pain)
- 3) Shortness of breath
- 4) Feeling feverish, fatigue, malaise



Signs (Doctor Finds):

Fever (↑ body temperature)

Fast breathing (tachypnea)

Decreased air entry on auscultation

Crackles or bronchial breath sounds

Dullness to percussion over affected area

Possible tachycardia (↑ heart rate)

Cardiovascular Example: Myocardial Infarction (Heart Attack)

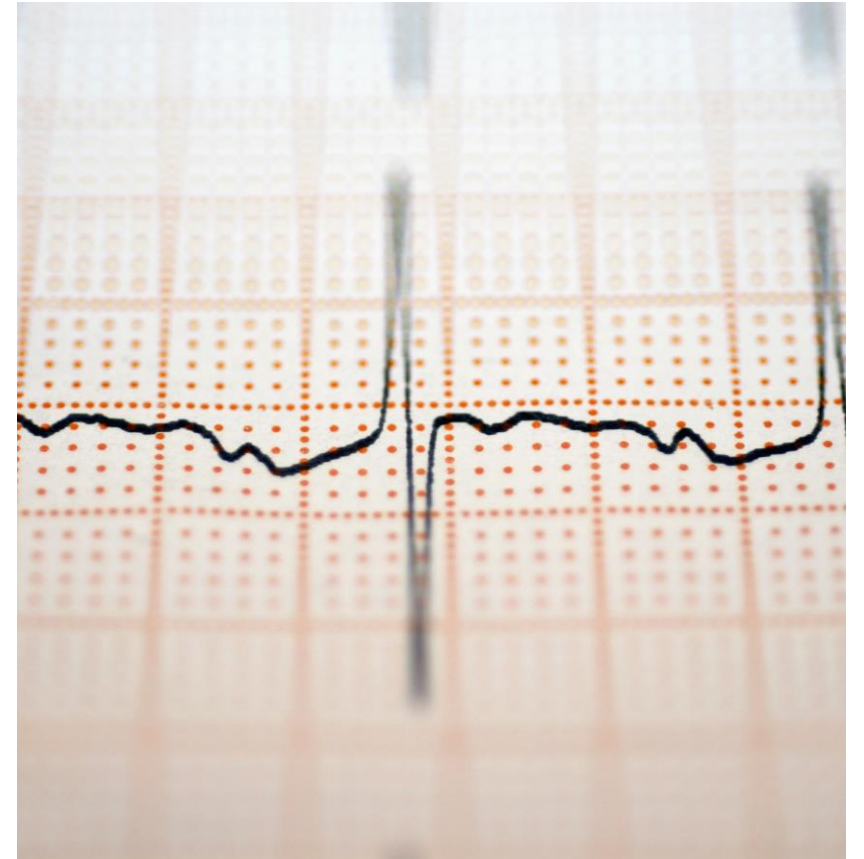
Symptoms: Severe chest pain or pressure, often central

Pain may radiate to left arm, jaw, neck, or back

Shortness of breath

Nausea, vomiting

Profuse sweating, feeling of impending doom



Signs:

Pale, cold, sweaty skin

- Abnormal heart rate (tachycardia or bradycardia)
- Hypotension in severe cases
- ECG changes (e.g. ST elevation)
- Raised cardiac enzymes (troponin) in blood tests

Endocrine Example: Diabetes Mellitus

Symptoms:

- Polyuria (frequent urination)
- Polydipsia (excessive thirst)
- Polyphagia (increased hunger)
- Unintentional weight loss
- Blurred vision
- Fatigue



Signs:

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graph LR; Signs[Signs:] --- Dehydration[Dehydration (dry mucous membranes, poor skin turgor)]; Signs --- Wound[Slow wound healing, recurrent infections]; Signs --- Sensation[Loss of sensation in feet (peripheral neuropathy)]; Signs --- Glucose[Elevated blood glucose level on testing];
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Dehydration (dry mucous membranes, poor skin turgor)

Slow wound healing, recurrent infections

Loss of sensation in feet (peripheral neuropathy)

Elevated blood glucose level on testing

Hematology Example: Iron Deficiency Anemia

Symptoms:

- Generalized fatigue, weakness
- Shortness of breath on exertion
- Palpitations
- Dizziness or light-headedness
- Headache
- Sometimes pica (craving for clay, ice, etc.)



Hematology Example: Iron Deficiency Anemia

Signs:

Pallor of skin, conjunctiva, palms

Brittle nails, koilonychia (spoon-shaped nails)

Glossitis (smooth, sore tongue)

Tachycardia

Systolic flow murmur

Surgical Example: Acute Appendicitis

Symptoms:

- Abdominal pain:
 - Initially vague, around the umbilicus
 - Later localized to right lower quadrant
- Nausea and vomiting
- Loss of appetite (anorexia)
- Mild feverish feeling



Surgical Example: Acute Appendicitis

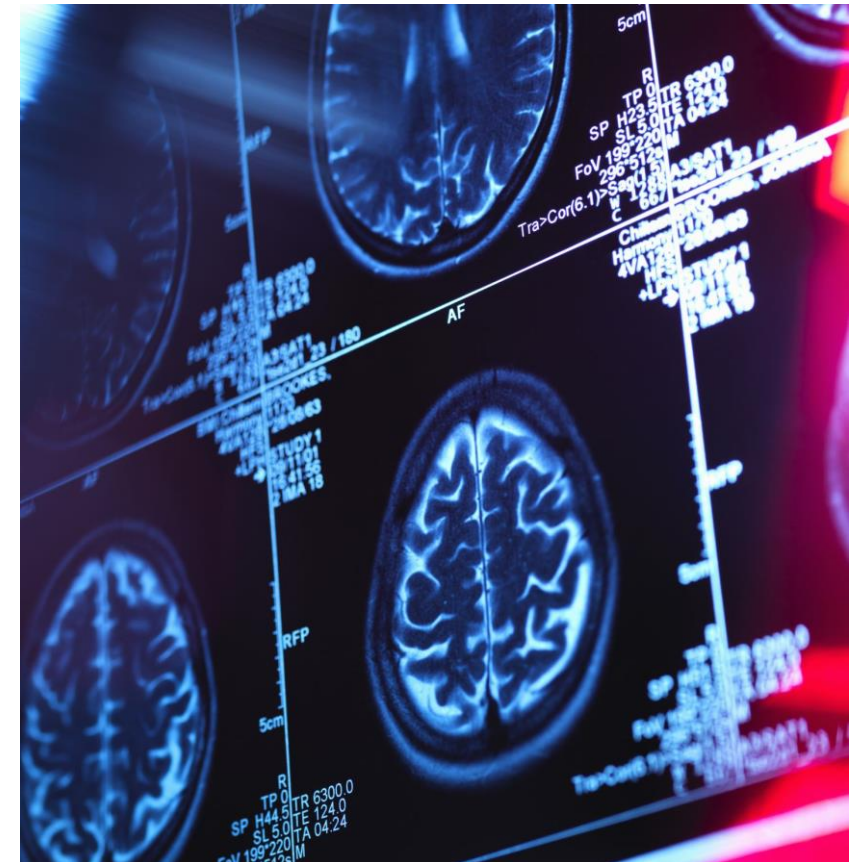
Signs:

- Tenderness at McBurney's point
- Guarding and rebound tenderness in right lower quadrant
- Low-grade fever
- Tachycardia
- Positive Rovsing's sign, psoas sign (optional for more advanced students)

Neurology Example: Stroke

Symptoms (sudden onset):

- Weakness or numbness on one side of the body
- Difficulty speaking or understanding speech
- Loss of vision in one eye or one visual field
- Sudden severe headache (especially in hemorrhagic stroke)
- Loss of balance or coordination



Signs:

Facial droop

Hemiparesis (weakness of
one side)

Slurred speech or aphasia

Asymmetrical reflexes,
Babinski sign

Gait disturbance, positive
neurological findings

Cardiac Example: Heart Failure

- **Symptoms:**
- Shortness of breath, especially on exertion
- Orthopnea (difficulty breathing when lying flat)
- Paroxysmal nocturnal dyspnea (waking up breathless at night)
- Fatigue, reduced exercise tolerance
- Swelling of legs



Signs:

Elevated jugular venous pressure (JVP)

Bilateral ankle edema

Crackles in lungs (pulmonary edema)

Enlarged liver (hepatomegaly)

S3 gallop on auscultation (in some patients)

Hepatic Example: Liver Cirrhosis

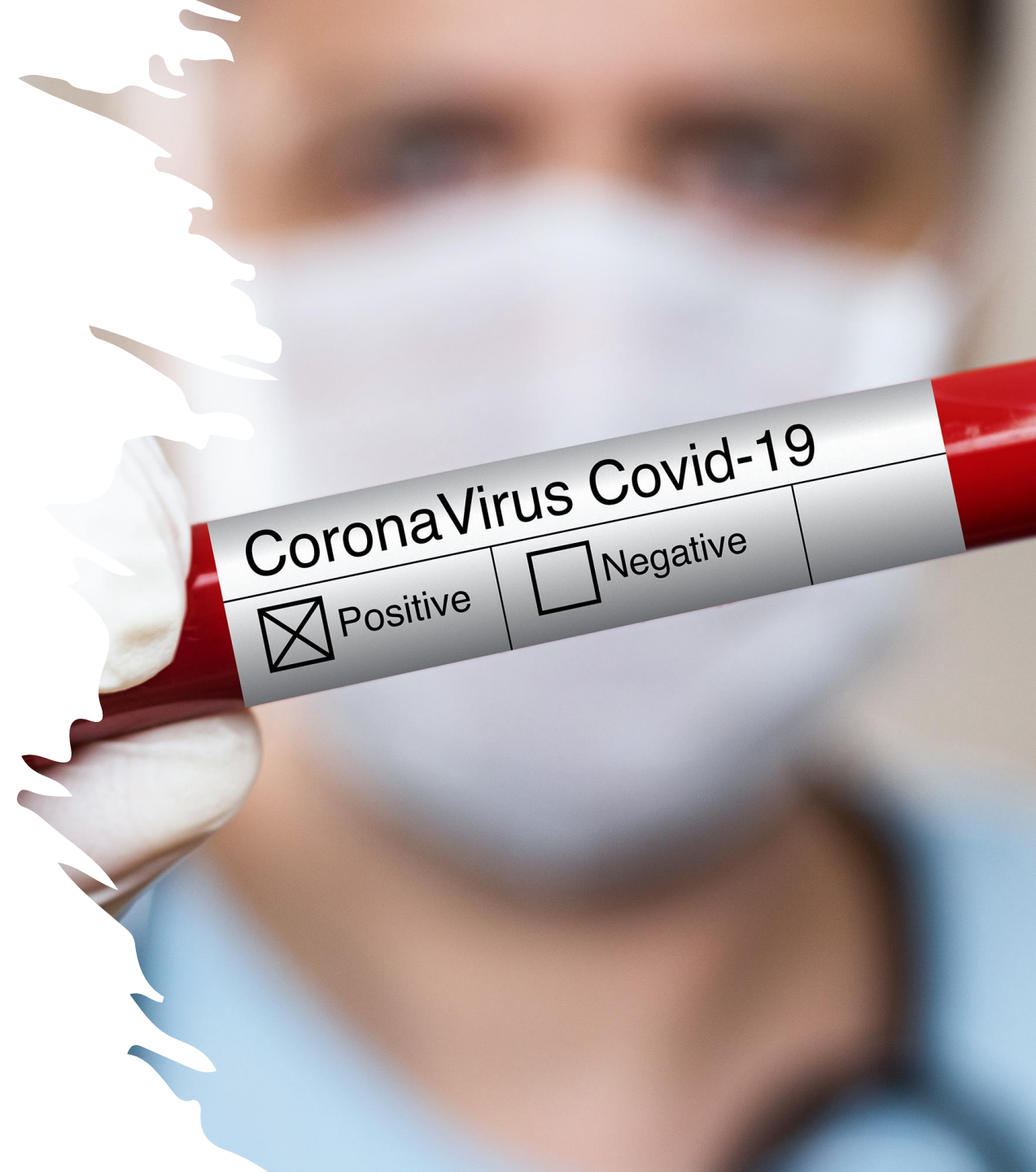
Symptoms:

Fatigue, weakness

Loss of appetite,
nausea

Abdominal distension
(feeling of fullness)

Itching (pruritus)



Hepatic Example: Liver Cirrhosis

Signs:

- 1) Jaundice (yellow discoloration of skin and eyes)
- 2) Ascites (fluid in the abdomen)
- 3) Spider angiomas, palmar erythema
- 4) Gynecomastia in males
- 5) Splenomegaly
- 6) Bruising due to coagulopathy



Urinary Example: Urinary
Tract Infection (UTI)

Symptoms:

- Dysuria (painful urination)
- Frequency and urgency of urination
- Suprapubic pain
- In pyelonephritis: flank pain, fever, chills, nausea

Urinary Example: Urinary Tract Infection (UTI)

Signs:

- Suprapubic tenderness
- Costovertebral angle tenderness (in pyelonephritis)
- Fever
- Positive urine dipstick (leukocytes, nitrites, blood)

Summary

Disease manifests through **signs** (objective) and **symptoms** (subjective).

Careful history and physical examination help link **patterns of signs and symptoms** to specific diseases.

Proper use of **medical terminology** improves communication and patient care.



Thank you